

Contact

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Top Skills

Project Management
advanced dsa
cs fundamental

Certifications

full stack development

mukesh prasad

Backend Engineer | Building Scalable APIs & Distributed Systems |
C++ | Python · FastAPI · Redis · PostgreSQL | NIT CALICUT
Bengaluru, Karnataka, India

Summary

I'm a Backend Engineer building scalable APIs, distributed systems, and backend infrastructure using Python, FastAPI, PostgreSQL, and Redis. Before writing code professionally, I spent 2 years as a Design Engineer at Kia Motors — working on suspension systems for Kia Seltos and Sonet. That engineering background gave me something most developers don't have: a systems-first mindset. I think in components, failure modes, and tradeoffs before I write a single line. Over the past two years I've built production-grade backend systems independently: Weather Alert Platform — FastAPI + Redis caching + PostgreSQL + Docker + scheduled alerting Distributed Rate Limiter — Token Bucket algorithm with Redis atomic ops Hotel Booking System — concurrency control with distributed locking LRU Cache, Ride Sharing, ATM System — C++ LLD systems with design patterns I'm actively looking for Backend Engineer / SDE roles in Bangalore. Open to conversations — reach me at mukeshprasadnitc929@gmail.com

Experience

Personal Backend Projects

Back End Developer

July 2025 - Present (1 year)

Bengaluru

- * Built a weather alert platform using FastAPI, PostgreSQL, Redis, and Docker with support for forecast retrieval, caching, and user-defined weather alerts.
- * Implemented Redis-based caching with TTL expiration to minimize external API dependency and improve application performance.
- * Developed a backend reservation system with concurrency control using atomic operations and distributed locking techniques.
- * Built a distributed rate limiter using Redis and the Token Bucket algorithm to protect APIs from excessive traffic.

* Designed reusable backend components including thread-safe caching, database persistence layers, and RESTful APIs following clean architecture principles.

Udemy

Python Instructor

June 2024 - Present (2 years 1 month)

BANGALORE

Aakash Educational Services Limited

Software Engineer

October 2020 - July 2025 (4 years 10 months)

BANGALORE

- Designed and built internal Python automation tools to streamline student performance tracking and batch reporting.
- Eliminated manual workflows used by 10+ faculty members daily.
- Developed data pipeline scripts integrating academic performance data with PostgreSQL, enabling structured analytics and reducing report generation time by ~70%.
- Built automated email notification system using Python (SMTP) to dispatch performance alerts, exam schedules, and result summaries to 1,000+ students per batch — replaced entirely manual communication process.
- Implemented scripted data extraction and transformation workflows to consolidate attendance, test scores, and batch metrics into structured PostgreSQL tables for reporting.
- Mentored 6,000+ students (1,000+ annually) in analytical problem-solving, structured reasoning, and exam strategy — developed curriculum frameworks built on algorithmic decomposition techniques.
- Designed structured problem-solving frameworks adopted across batches — directly sharpened own DSA and systems thinking skills now applied to backend engineering.

Kia Worldwide

DESIGN ENGINEER

April 2018 - August 2020 (2 years 5 months)

Chennai

- Contributed to design and validation of suspension systems used in Kia Seltos and Kia Sonet, improving vehicle stability and ride comfort

Education

National Institute of Technology Calicut

Bachelor's degree, Mechanical Engineering · (August 2014 - April 2018)